

MySQL 跨服务器远程访问环境搭建与排错记录

一、环境概述

项目	信息
目的	模拟企业真实环境：两台服务器，客户端访问远程 MySQL 数据库
服务器端	CentOS Stream 9 / IP: 192.168.56.110 / MySQL 8.0
客户端	Ubuntu 24.04 / IP: 192.168.56.10
虚拟化软件	VirtualBox / VMware (Host-Only + NAT)

二、网络拓扑



三、服务器端（CentOS）搭建步骤

3.1 安装 MySQL

```
sudo dnf install mysql-server -y
sudo systemctl start mysqld
sudo systemctl enable mysqld
```

3.2 获取临时密码

```
sudo grep 'temporary password' /var/log/mysqld.log
```

3.3 修改密码

```
sudo mysql -u root -p
# 输入临时密码
ALTER USER 'root'@'localhost' IDENTIFIED BY 'YourStrong@123';
FLUSH PRIVILEGES;
```

四、配置 MySQL 允许远程连接

4.1 修改监听地址

```
sudo vi /etc/my.cnf
```

在 [mysqld] 段落添加：

```
bind-address = 0.0.0.0
```

```
bind-address = 0.0.0.0
mysqlx-bind-address = 0.0.0.0
~
```

4.2 重启 MySQL

```
sudo systemctl restart mysqld
```

4.3 验证监听端口

```
sudo netstat -tunlp | grep 3306
```

```
[root@node1 ~]# netstat -tunlp | grep 3306
tcp        0      0 0.0.0.0:3306 0.0.0.0:*        LISTEN    18466/mysqld
tcp        0      0 0.0.0.0:33060 0.0.0.0:*        LISTEN    18466/mysqld
```

4.4 开放防火墙端口

```
sudo systemctl stop firewalld    #关闭防火墙
sudo systemctl status firewalld  #查看防火墙状态
```

```
[root@node1 ~]# systemctl status firewalld.service
○ firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; disabled; preset: enabled)
   Active: inactive (dead)
     Docs: man:firewalld(1)
[root@node1 ~]#
```

五、创建远程访问用户

5.1 登录数据库

```
mysql -uroot -p    #登录 root 账户
```

```
[root@node1 ~]# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

5.2 创建访问用户

```
CREATE USER 'remote'@'%' IDENTIFIED BY 'Remote@123456'; #用户名为 remote
GRANT ALL PRIVILEGES ON *.* TO 'remote'@'%';          #授予最高权限
FLUSH PRIVILEGES; #刷新权限
```

验证用户创建成功：

```
SELECT User, Host FROM mysql.user WHERE User='remote';
```

```
mysql> SELECT User, Host FROM mysql.user WHERE User='remote';
+-----+-----+
| User   | Host |
+-----+-----+
| remote | %    |
+-----+-----+
1 row in set (0.00 sec)
```

六、客户端（Ubuntu）连接测试

6.1 安装 MySQL 客户端

```
sudo apt update
sudo apt install mysql-client -y
```

6.2 连接远程数据库

```
mysql -h 192.168.56.110 -u remote -p
# 输入密码: Remote@123456
```

成功连接输出：Welcome to the MySQL monitor. ...

mysql>

```
root@Ubuntu:~# mysql -h 192.168.56.110 -u remote -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql>
```

至此，我们的 MySQL 跨服务器远程访问服务部署完成，下面我们看一下在部署过程中遇到的错误

七、遇到的错误及解决方案

错误 1: ERROR 1290 (HY000) --skip-grant-tables

现象：

```
The MySQL server is running with the --skip-grant-tables option
so it cannot execute this statement
```

```
mysql> CREATE USER 'remote'@'%' IDENTIFIED BY '123456';
ERROR 1290 (HY000): The MySQL server is running with the --skip-grant-tables option
so it cannot execute this statement
mysql> GRANT ALL PRIVILEGES ON *.* TO 'remote'@'%';
```

原因：MySQL 启动时跳过了权限表加载。

解决方法：

```
# 搜索配置文件中的 skip-grant-tables
```

```
sudo grep -r "skip-grant-tables" /etc/my.cnf /etc/my.cnf.d/
```

```
[root@node1 my.cnf.d]# grep -r "skip-grant-tables" /etc/my.cnf /etc/my.cnf.d/
/etc/my.cnf:skip-grant-tables
```

设置这个跳过权限可以实现免密登录，但是在创建用户会出现错误，建议不要免密登录

```
# 编辑/etc/my.cnf，删除或注释掉该行
```

```
sudo vi /etc/my.cnf
```

```
[mysqld]
```

```
#免密登录
```

```
#skip-grant-tables
```

```
# skip-grant-tables ← 删除或加#
```

```
sudo systemctl restart mysqld
```

错误 2: ERROR 1819 (HY000) - Password policy

现象：

```
Your password does not satisfy the current policy requirements
```

```
mysql> CREATE USER 'remote'@'%' IDENTIFIED BY '123456';
ERROR 1819 (HY000): Your password does not satisfy the current policy requirements
```

原因：MySQL 默认密码策略要求大小写+数字+特殊字符。

解决方案 A（推荐）：使用符合策略的密码

```
CREATE USER 'remote'@'%' IDENTIFIED BY 'Remote@123456';
```

解决方案 B（测试环境）：降低密码策略等级

SET GLOBAL validate_password.policy = LOW;	#安全等级较低
SET GLOBAL validate_password.length = 4;	#密码长度至少 4 位

我这里使用的方案 A

错误 3：ERROR 2003 (HY000) - Can't connect to MySQL server

现象：

Can't connect to MySQL server on '192.168.56.110:3306' (111)
<pre>mysql> show databases; ERROR 2013 (HY000): Lost connection to MySQL server during query No connection. Trying to reconnect... ERROR 2003 (HY000): Can't connect to MySQL server on '192.168.56.110:3306' (111) ERROR: Can't connect to the server</pre>

排查步骤：

步骤	命令	目的
1	ping 192.168.56.110	检查网络连通性
2	sudo systemctl status mysql	检查 mysql 服务状态
3	sudo netstat -tunlp grep 3306	确认监听 0.0.0.0 而非 127.0.0.1
4	sudo firewall-cmd --list-all	检查防火墙是否开放 3306

解决：修改 `bind-address = 0.0.0.0` 并重启 MySQL。

九、经验总结

1. 远程连接三大要素：

- MySQL 监听 0.0.0.0
- 防火墙开放 3306 端口
- 用户 host 为 %

2. 排查思路按层递进：

- 网络层: ping
- 端口层: telnet
- 服务层: systemctl status
- 权限层: SELECT User, Host

3. 小贴士:

- CentOS 用 firewall-cmd, Ubuntu 用 ufw
- 遇到错误先看日志: /var/log/mysqld.log
- 生产环境不要用 remote@% 这种宽松权限

十、后续学习建议

- 配置 MySQL 主从复制
- 用 Docker Compose 搭建 MySQL 集群
- 学习 MySQL 慢查询分析与优化
- 编写 Shell 脚本实现自动备份